

CAUE603 Data Analytics

MCA – III

Academic Year-2025-26

LAB Task - 2: Learn Python Numpy

NUMPY CONCEPTS

NumPy is a Python library used for **numerical** and **matrix computations**.

NumPy Features:

- **Fast Array Processing**
- NumPy arrays are **more efficient** than Python lists for large data.
- **Vectorized Operations**
- Perform **element-wise** operations **without** writing **loops**.
- **Multidimensional Arrays**
- Support for **arrays of any dimension**.
- **Integration**
- Works with other **libraries** like **Pandas**, **Matplotlib**, and **Scikit-learn**
- We have list in python but NumPy provides array object that is much faster than lists
- Array object of NumPy is known as “ndarray”.
- We can create “ndarray” object using “array()” function.
- NumPy arrays are stored at one continuous place in memory unlike lists, so processes can access and manipulate them very efficiently. (locality of reference concept)

NUMPY COMMANDS

Function	Name of command	Python code
Creating array/vector	Np.array	import numpy as np ages = [16, 22, 39, 86] arr = np.array(ages) print(arr)
Create array fill with zero	Np.zeros	import numpy as np ages = np.zeros((3,2)) arr = np.array(ages) print(arr)
Create array fill with one	Np.ones	import numpy as np ages = np.ones((3,2)) arr = np.array(ages) print(arr)

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Create array with start, stop, increment values	Np.arange	import numpy as np ar = np.arange(5, 25, 5) print(ar)
Array attributes	Shape Dtype Size ndim	import numpy as np n1 = np.arange(5, 25, 5) print(n1) print(n1.size) print(n1.dtype) print(n1.ndim) ----- import numpy as np arr1=np.array([[1,2,3],[4,5,6],[7,8,9]]) print(arr1) print(arr1.shape)
Basic operations Addition Subtraction Division Multiplication Exponential	+ - / * **	import numpy as np n1 = np.arange(5, 25, 5) n2=[1,2,3,4] print(n1+n2)
Aggregate and statistical functions	Np.Sum Np.Min Np.Max Np.mean Np.std Np.average Np.median	import numpy as np n1 = np.arange(5, 25, 5) n2=[1,2,3,4] print(np.max(n1)) print(np.sum(n2)) print(np.mean(n1)) print(np.median(n1)) print(np.std(n2)) ----- import numpy as np n1= np.random.randint(3,6,10) print(n1.max()) import numpy as np marks = np.array([60,70,70,60,23,100]) avg = (round(np.average(marks),2)) print(avg)
Reshaping array	Reshape()	import numpy as np

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Print horizontal/vertical		<pre>mks=[10,29,22] n1 = np.array(mks) n2=n1.reshape((3,1)) print(n2)</pre>
Combine multiple arrays to one	np.concatenate()	<pre>import numpy as np mks=[10,29] n1 = np.array(mks) n2=np.arange(5,25,10) print(n2) n3=np.concatenate((n1,n2)) print("merged=", n3)</pre> <pre>import numpy as np arr1=np.array([[1,2,3],[1,2,3]]) arr2=np.array([[4,5,6],[4,5,6]]) print(np.concatenate((arr1,arr2),axis=1)) #joining two arrays along rows</pre>
Splitting array	Np.array_split	<pre>import numpy as np arr1=np.array([1,2,3,4,5,6,7,8,9,10]) print(np.array_split(arr1,4)) #splitting array into 4 pieces</pre> <pre>import numpy as np arr1=np.array([1,2,3,4,5,6,7,8,9,10]) arr2=np.array_split(arr1,5) print(arr2[0]) print(arr2[1]) print(arr2[2]) print(arr2[3]) print(arr2[4])</pre>
Slicing of Array	lower:upperlimit	<pre>import numpy as np arr1=np.array([1,2,3,4,5,6,7,8,9]) print(arr1[1:4])</pre>
Generating Random number	Random.rand	<pre># Generate and print random float import numpy as np print(np.random.rand())</pre> <pre>import numpy as np N=5 salary = 5000+(8000-2000)* np.random.rand(N) print(salary)</pre>

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	Random.randn Random.randint	import numpy as np print(np.random.randn(1,2)) # Generate and print random number import numpy as np print(np.random.randint(3,6,10))
Permutation and Combination of Values to be printed	Np.random.choice	#Randomly print the values import numpy as np N=5 gen = np.random.choice(['Male', 'Female'], N, p=[0.6, 0.4]) print(gen)
Searching value in array	Np.where()	import numpy as np arr1=np.array([1,2,3,1,2,3,1,2,3]) print(np.where(arr1 == 3)) #gives all indexes where value 3 is found import numpy as np arr1=np.array([1,2,3,4,5,6,7,8,9,10]) print(np.where(arr1%2 == 0)) #gives all indexes where value is even
Sorting into array	Np.sort()	import numpy as np arr1=np.array([2,3,1,4,3,5,8,6]) print(np.sort(arr1))
Reverse of the array	Np.flipud()	import numpy as np arr1=np.array([2,3,1,4,3,5,8,6]) print(arr1) print(np.flipud(arr1)) #reversing the array import numpy as np arr1=np.array([1,2,3,4,5]) print(arr1) print(np.flip(arr1)) #flipping the 1D array
USER INPUT Create array from user	Np.array Np.append Input print	import numpy as np arr1= np.array([]) a=int(input("Enter the size of array")) for i in range(a):

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		<pre>v=int(input("Enter element :")) arr1= np.append(arr1, v) print(arr1)</pre>
MATRIX CREATION 2X2 3X3	Np.array	<pre>import numpy as np # create a 2x2 matrix matrix1 = np.array([[1, 3], [5, 7]]) print("2x2 Matrix:\n",matrix1) # create a 3x3 matrix matrix2 = np.array([[2, 3, 5], [7, 14, 21],[1, 3, 5]]) print("\n3x3 Matrix:\n",matrix2)</pre>
Matrix Manipulation Addition Subtraction Multiplication Division Transpose Inverse	+ - * / Np.transpose Np.linalg.inv	<pre>import numpy as np # create a 2x2 matrix m1 = np.array([[1, 3], [5, 7]]) print("Matrix 1:\n",m1) m2 = np.array([[2, 3], [7, 14]]) print("\n Matrix 2:\n",m2) # Finding product of two matrix s=m1*m2 print("sum=") print(s)</pre> <pre>import numpy as np # create a 2x2 matrix m1 = np.array([[1, 3], [5, 7]]) print(np.transpose(m1))</pre> <pre>import numpy as np # create a 2x2 matrix m1 = np.array([[1, 3], [5, 7]]) print(np.linalg.inv(m1))</pre>
multi-dimensional iterator object to iterate over arrays	np.nditer	<pre>#iteration using "nditer" import numpy as np arr1=np.array([[[1,2,3],[4,5,6]],[[7,8,9],[10,11,12]]]) for k in np.nditer(arr1): print(k,end=" ")</pre>
Slicing and Dicing in matrix	:	<pre>import numpy as np n_array = np.array([[1, 2, 3],</pre>

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		<pre> [4, 5, 6], [7, 8, 9]]) #(element at first row, second column) : print(n_array[0, 1]) #array([1, 2, 3]) (first row, all columns) : print(n_array[0, 0:3]) # (first row, all columns): print(n_array[0, :]) #(all rows, second column): print(n_array[:, 1]) array_2d = np.array([[0, 1, 2, 3], [4, 5, 6, 7], [8, 9, 10, 11], [12, 13, 14, 15]]) # Get the first two rows and the first three columns diced_array = array_2d[:2, :3] print(diced_array) </pre>
Permutation and Combination of Values to be printed	Np.choice	<pre> #Randomly print the values import numpy as np N=5 gen = np.random.choice(['Male', 'Female'], N, p=[0.6, 0.4]) print(gen) </pre>